🧠 AI Engineer – Take-Home Challenge

About Jobnova:

Jobnova is an AI agent–driven future job infrastructure that automates how people discover, match, and apply for work. We believe humans deserve better opportunities in the AI era!

AI agents help us find direct hiring signals and we update on our [Linkedin page](#).

About this take-home challenge

AI coding is very encouraged, and there are 3 parts of the challenge, you can select 1--2 that you are mostly familiar with to implement a demo.

Once you finish, send your submission and a go-through video to kevin@libaspace.com

Note: Each part is the real product feature we are going to build, so please be serious on your demo.

Part 1

AI Mock Interview demo

1. Implement the self-introduction and past-experience stages of the interview process using LiveKit AI Multi-Agent, ensuring stable performance and clear interaction logic.
2. Ensure smooth and natural transitions between the two stages, with well-defined switching logic and no interruptions, conflicts, or repetitive prompts.
3. Provide a time-based fallback mechanism for stage transitions to guarantee continuous workflow progression when the normal switching logic is not triggered.
4. https://github.com/livekit/agents

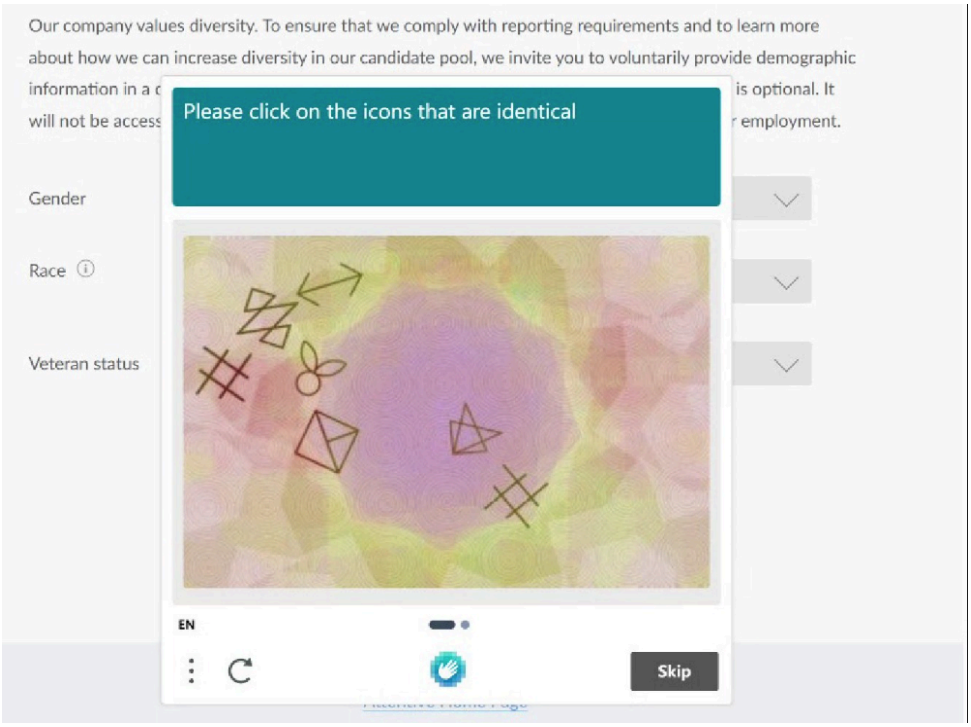
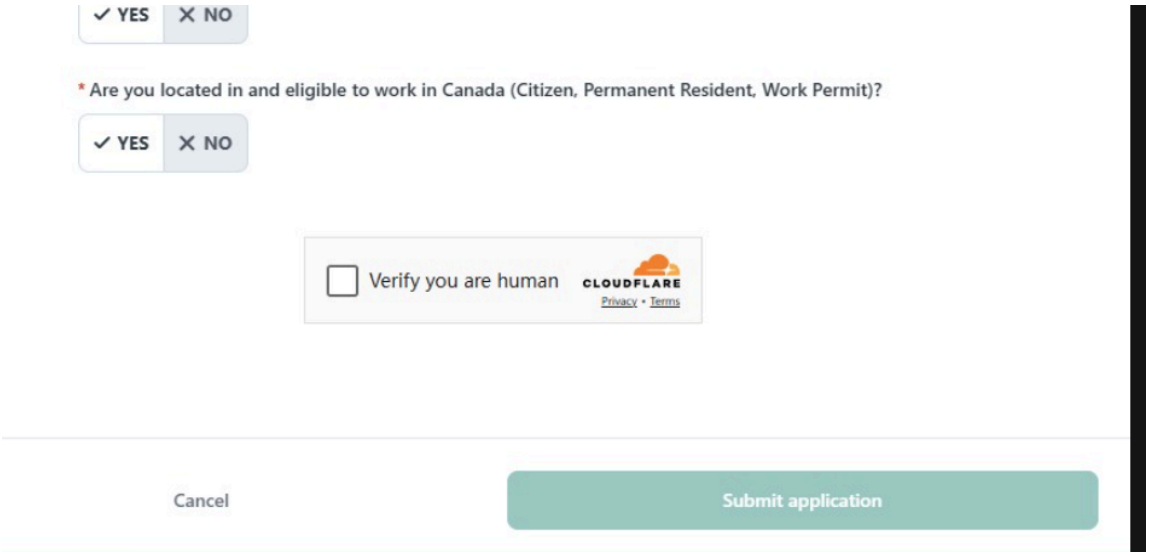
Part 2

AI Job Source agent

1. From Linkedin job listing pages, crawl the company name and company website URL(You may utilize third party Linkedin crawler API if you need)
2. From company website URL,use web agent to navigate to the career page URL
3. From the career page, get one opening position's URL
4. Return the results in format: company name,career page URL,open position's URL

Part 3

Resume auto apply agent

1. While submitting job applications is time consuming, we are developing a job auto apply agent for ATS based websites like greenhouse workday Glassdoor ... etc.
2. For now we use playwright to fill the application forms in the headless chrome in EC2 server. But we will meet human verification when the playwright tries to click the submit button for some website, sometimes like workday, lever.
3. We would need to build a chrome plug-in, and run it purely on server side, so it can work around with this.
4. Your task: Work on the following job application from Lever, that can auto apply without the blocking issue(consider chrome extension to automate):

https://jobs.lever.co/ekimetrics/d9d64766-3d42-4ba9-94d4-f74cdaf20065